

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (Mechanical) (Credit System) Degree Examinations

May - June 2023

20ME602 - DESIGN OF MACHINE ELEMENTS - II

Duration: 3 Hours

Max. Marks: 100

- Note: 1) Answer **Five full** questions choosing **One full** question from **each Unit**.
 2) Use of Design **Data Hand book** is permitted.
 3) Missing data, if any, may be suitably assumed.

Unit - I

Marks BT* CO* PO*

1. a) Determine the dimensions of I-section, as shown in Fig. 1a, in which maximum fiber stresses are numerically equal where $b_i + b_o = 120$ mm.

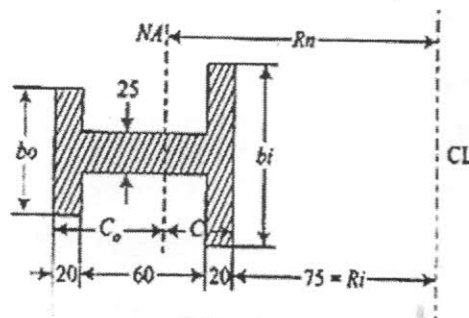


Fig. 1a

12 L*3 1 1

- b) Determine the maximum tensile, compressive and shear stresses induced in a beam loaded as shown in Fig. 1b.

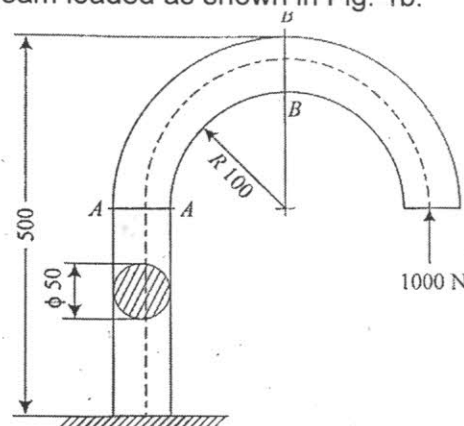


Fig. 1b.

8 L3 1 2

2. a) Fig. 2a shows a punch press is subjected to pure bending in its plane of symmetry. Find the proper value of thickness t so that extreme fiber stress in pure bending is numerically equal. If this section is subjected to a bending moment of 3 kN-m, determine the maximum fiber stress induced.

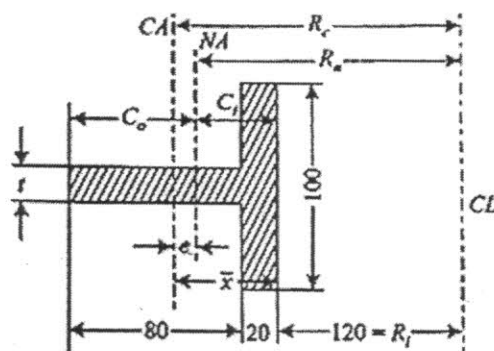


Fig. 2a

12 L3 1 1

- b) A chain link is made of 30 mm diameter steel rod has a mean diameter of 90 mm. The straight sides are each 60 mm long. If the link carries a load of 9 kN, determine the maximum fiber stresses induced in the link.

08 L3 1 2

Unit – II

3. a) A spring loaded safety valve for a boiler required to blow off at a pressure of 0.8 MPa. The diameter of the valve seat is 90 mm and the maximum lift is of 10 mm. Design a suitable spring for the valve assuming the spring index as 7. Provide an initial compression of 30 mm. Take allowable shear stress as 420 MPa. Assume squared and ground ends.
- b) A loaded narrow gauge car weighing 1800 kg and moving at a speed of 72 m/min is brought to rest by a bumper consisting of 2 helical steel springs. In bringing the car to rest the springs undergo a compression of 200 mm. The allowable shear stress in the spring material is 270 MPa. Spring index is 6 and modulus of rigidity is 84 GPa. Design the suitable spring. Assume squared and ground ends.
4. a) A truck spring has 3 full length and 9 graduated leaves. The span is 1.1 m and width of central band is 150 mm. The load on spring is 9000 N. The allowable stress is 500 MPa in all the leaves. The deflection under load is 72 mm. Design the spring. Assume $E=206$ GPa.
- b) A car weighing 10000 N without passengers has a seating capacity of 4 passengers each weighing 1000 N. The wheel base is 2.2 m and cg of the loaded car is 1.2 m behind front axle. The car is to be supported on four semi elliptical carriage springs of each 900 mm between the supports. Design a suitable spring using FoS as 1.8 based on the proof stress of 810 MPa. Static loads are multiplied with a factor of 2.2 to allow the impacts. The spring should have a deflection of 60 mm. Neglect central load, $E=206$ GPa, Number of full length leaves 2 and total leaves are 8.

08 L3 2 1

12 L3 2 2

10 L3 2 2

10 L3 2 2

Unit – III

5. a) An asbestos lined cone clutch is to transmit a maximum torque of 100 Nm. The maximum axial force is 850 N. Find the required width of the friction lining. Total cone angle is 24° . Maximum allowable pressure on the friction lining is 0.1 MPa. Assume coefficient of friction is 0.2.
- b) A double shoe brake as shown in Fig. 5b is capable of absorbing a torque of 1.5 kN-m. The diameter of brake drum is 400 mm. Angle of contact of each shoe is 110° . The coefficient of friction is 0.4. Design the brake and find the spring force necessary to set the brake. Assume normal pressure 0.2 MPa. Also design the shoe and lever by assuming the depth twice the width. Take bending stress for the lever 117.68 MPa.

08 L3 3 2

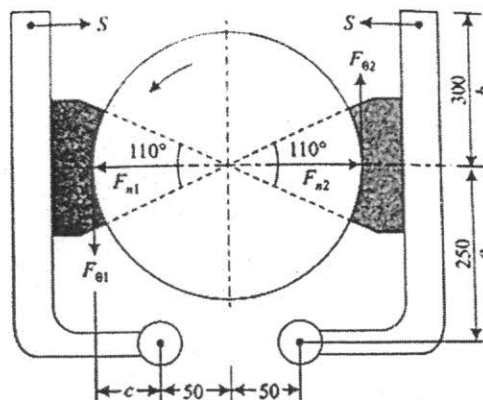


Fig 5b

12 L3 3 2

6. a) A multiplate clutch with steel and bronze plates is to transmit 18 kW at 1440 rpm. The inner and outer diameters of contact are 80 mm and 140 mm respectively. The coefficient of friction is 0.1 and allowable pressure is 0.35 MPa. Assume uniform wear and determine the number of steel and bronze plates. 10 L3 3 2
- b) For a simple band brake with one end passing through the lever fulcrum, determine the torque capacity of an operating force of 300 N at the end of the lever. The length of the lever is 900 mm and lever arm is 600 mm. The angle of contact of band over the drum is 240°. Coefficient of friction is 0.21. The drum rotates at 200 rpm and has a diameter of 360 mm. 10 L3 3 2

Unit – IV

7. Design a pair of spur gear to transmit 24 kW at 1000 rpm to a parallel shaft to be rotated at 400 rpm. The centre distance between shafts is 175 mm. Assume 20° FDI, Material for pinion and gear is untreated cast steel, service factor 1.5, Face width is 10 times the module. 20 L3 4 2
8. Design a pair of helical gear to transmit 15 kW at 1200 rpm of the pinion. The gear is to rotate at 600 rpm. The helix angle is 17.5°. The center distance between the gears is 150 mm. The pinion is made up of high carbon steel with static stress 103.5 MPa and gear of 0.4% Carbon steel untreated with static stress 69.6 MPa. Assume 20° FDI, Face width is 10 times the normal module. Assume scant lubrication and service factor 1.5. 20 L3 4 2

Unit – V

9. Design the main bearings of 4-stroke diesel engine to sustain a load of 6 kN. The operating speed of the shaft is 100 rpm. 20 L3 5 1
10. a) Derive Petroff's equation of coefficient of friction 08 L2 5 1
- b) Select a suitable ball bearing required to be mounted on a shaft of diameter 45 mm to withstand a radial load of 6 kN and thrust load 3 kN at a rated speed of 300 rpm. The bearing works for 50 hours per week for 3 years. Assume light shock with service factor 1.25. 12 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
